RELATIONAL THEORY OF EXPRESSION

Quasilocal Manifestation, Reconstruction, and Operational Equivalence

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> **Motif.** Physics does not possess reality as an unconditioned object. It studies the conditions under which physical content becomes expressible in quasilocal relations.

Status

This document presents a mathematical research programme, not an established physical theory. Definitions, propositions, postulates, hypotheses, interpretations, and open problems are kept distinct throughout.

Abstract

Relational Theory of Expression (RTE) is formulated as a mathematical and methodological framework for quasilocal physical expression. Its central claim is neither that reality is created by observation nor that models are arbitrary instruments. Rather, the physically available content in a domain is determined through a joint configuration of causal-geometric domain, observer interface, admissible operations, and formal reconstruction. The first strict core is a domain-indexed expression map or relation E_D. It induces an operational equivalence relation on candidate physical states: two states are equivalent when no admissible expression in the domain distinguishes them. The corresponding quotient class [x]_D is the primary object available to that domain. Local expression data are organized presheaf-wise; gluing and obstruction distinguish local compatibility from global realizability. Reconstruction of past and future is treated as an inverse problem yielding classes or posterior distributions of compatible histories, not direct possession of unobserved epochs. An auxiliary B^4 filling is used as a worked example of type discipline: mathematical existence as a filling or variational support does not entail physical existence as an exterior volume. A compatible information layer is supplied by carrier-dependent distinguishability. The result is a compact formal nucleus for RTE and a controlled route toward later dynamical, cosmological, and perceptual applications.

Keywords: relational ontology; quasilocal physics; operational equivalence; expression maps; presheaves; reconstruction; inverse problems; compact S^3 cosmology; B^4 filling; observer-dependent information.

Claim-status convention

Label	Meaning
Definition	A stipulation of vocabulary inside RTE.
Theorem / Proposition / Lemma	A mathematical statement proved under stated assumptions.
Postulate	A physical or operational assumption linking the formalism to measurement.
Hypothesis	A proposed law-like or model-selecting principle not yet derived.
Interpretation	A philosophical or methodological reading; it proves no physics.
Open problem	A required construction or test not completed in Version 1.0.

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1. Position and Scope

RTE is not introduced here as a new dynamical law. It is a discipline for stating what kind of object a physical claim concerns when the claim is made from a quasilocal domain with finite causal access, finite resolution, and a specified reconstruction procedure.

A weak realism says that an object exists independently and the model represents it. Instrumentalism may set aside the object and retain only successful prediction. RTE rejects both reductions. It neither identifies model and reality nor detaches them. A model is treated as a constrained structure of expression: it preserves some relations, suppresses others, and must make the loss explicit.

> **Central thesis 1.1.** Physics does not directly describe self-subsistent things. It describes the conditions under which physical structures become expressible in quasilocal relations.

The present paper constructs only the first formal layer: domain-indexed expression, operational equivalence, local-to-global compatibility, and constrained reconstruction. It does not derive consciousness, quantum gravity, or dark-sector dynamics.

2. Primitive Structure: Expressibility Before Objects

The distinctive proposal is to place expressibility logically before the operational object. One does not begin with a completed object and then ask how it is represented. One begins with a compatibility structure under which a manifestation can occur. The operational object is then derived as an equivalence class of candidate realizations.

> **Primitive notion 2.1.** An expression condition is a structured compatibility relation among a candidate ground, a causal-geometric domain, an observer or measuring interface, a formal model, and an accessible manifestation.

$$\mathcal{E}_D(R, C_D, O, M ; P_D) = \text{true}.$$

The predicate form is intentionally prior to the map form. It accommodates nondeterminism, gauge redundancy, inverse problems, and cases in which several manifestations are compatible with the same data. A deterministic expression map is a special case.

$$E_D : S_phys \to S_obs(D).$$

The order of construction is therefore:

$$\text{conditions of expressibility} \to \text{manifestation} \to \text{operational object} \to \text{mathematical reconstruction}.$$

This reverses the usual order in which objects are assumed first and relations are added later.

3. Typed Ontology

The symbols used by RTE belong to different types. Confusion begins when they are treated as members of one undifferentiated ontology.

$$R : \text{GroundCandidate}$$

$$C_D : \text{CausalGeometricDomain}$$

$$O : \text{ObserverStructure}$$

$$M : \text{FormalModel}$$

$$P_D : \text{QuasilocalManifestation}$$

The ground candidate R is not assumed to be a hidden classical object with a complete catalogue of determinate properties. In stricter realizations it may be replaced by a class of compatible global sections, a state on an observable algebra, a limit object, a stack of local realizations, or a posterior family of histories.

Definition 3.1. A manifestation P_D is the physically accessible content licensed by an expression condition in domain D. It is not identical with the ground candidate, the observer, or the model.

Proposition 3.2. A sentence of the form "B^4 exists mathematically but is not physically postulated" is classically consistent, because mathematical existence and physical existence are distinct predicates.

Formally, Exists_Math(B^4) ∧ ¬Exists_Phys(B^4) has the form P ∧ ¬Q, not P ∧ ¬P.

4. Quasilocal Arena and Domains

For the cosmological branch, let the compact spatial arena be a three-sphere $\Sigma \simeq S^3$ with round scale A(t). The associated closed FLRW spacetime is

$$\mathcal{M} = I \times S^3_A, \quad ds^2 = -N^2(t)c^2dt^2 + A^2(t)d\Omega^{32}.$$

For a round three-sphere,

$$V_3(A) = 2\pi^2A^3, \quad {}^{(3)}R = 6/A^2.$$

Elementary separation 4.1. The absence of an exterior boundary does not imply absence of internal volume: ∂Σ = ∅ is compatible with Vol(Σ) > 0.

Definition 4.2. A quasilocal domain D is a physically distinguishable region D ⊂ Σ, usually with nonempty internal boundary ∂D, equipped with the metric, causal, and measurement data required to define accessible observables.

Different domains may induce different observable algebras, expression maps, equivalence relations, and reconstruction spaces. A domain is therefore not merely a subset of coordinates; it is an indexed physical access structure.

5. Expression Maps and Operational Equivalence

Definition 5.1. For a domain D, an expression relation is a relation $\mathcal{E}$_D ⊂ S_phys × S_obs(D). A deterministic expression map is the special case E_D : S_phys → S_obs(D).

Definition 5.2. For a deterministic expression map, define $x_1 \sim_D x_2$ if and only if $E_D(x_1) = E_D(x_2)$.

Theorem 5.3. The relation ~_D is an equivalence relation on S_phys.

Proof. Reflexivity, symmetry, and transitivity follow immediately from the corresponding properties of equality in S_obs(D).

Definition 5.4. The domain object accessible to D is the equivalence class [x]_D = {y ∈ S_phys : E_D(y) = E_D(x)}.

$$[x]_D = E_D^{-1}(E_D(x)).$$

Proposition 5.5 (quotient factorization). Let π_D : S_phys → S_phys / ~_D be the quotient map. Then E_D factors uniquely as E_D = ē_D ∘ π_D, where ē_D([x]_D) = E_D(x), and ē_D is injective onto im(E_D).

Proof. If [x]_D = [y]_D, then E_D(x) = E_D(y), so ē_D is well defined. Uniqueness follows because every state maps to its equivalence class. Injectivity onto the image is immediate.

This factorization is the first strict core of RTE. The accessible object is not an unconditioned representative x but a quotient object determined by the declared access structure.

5.1 Probabilistic and quantum variants

$$E_D: S_{phys} \to \mathrm{Prob}(Y_D).$$

Two candidate states are operationally equivalent when their induced probability measures agree for all admitted measurements. In a quantum realization one may use a channel or instrument

$$\mathcal{E}_D : \mathrm{States}(\mathcal{A}) \to \mathrm{States}(\mathcal{B}_D),$$

$$p(k \mid x, D, O, M) = \mathrm{Tr}[\mathcal{E}_k^D(\rho_x)].$$

Equality must be equality of all accessible outcome statistics, not merely equality of one recorded outcome.

6. Local-to-Global Organization

Let Open(Σ) denote a category of admissible domains. Assign to each domain D a set, space, algebra, or category $\mathcal{F}(D)$ of local expression data, with restriction maps for inclusions $U \subset D$.

$$\rho_{DU}: \mathcal{F}(D) \to \mathcal{F}(U).$$

> **Definition 6.1.** A local expression presheaf is a contravariant assignment $D \mapsto \mathcal{F}(D)$ satisfying identity and composition for restrictions.
>
> **Definition 6.2.** For a cover $\{D_i\}$ of D, local sections $s_i \in \mathcal{F}(D_i)$ are compatible when $s_i|_{D_i \cap D_j} = s_j|_{D_i \cap D_j}$ for every i,j.
>
> **Definition 6.3.** A compatible family is gluable if there exists $s \in \mathcal{F}(D)$ whose restriction to every D_i is s_i.
>
> **Proposition 6.4.** If $\mathcal{F}$ is a sheaf, every compatible local family glues uniquely. If $\mathcal{F}$ is only a presheaf, existence or uniqueness may fail.

Compatibility on overlaps must not be confused with global realizability. This distinction repairs a common error: local agreement alone does not prove the existence of a global physical state unless an appropriate gluing axiom or representability theorem is supplied.

6.1 Obstruction and contextuality

An obstruction is a mathematically specified failure of global realization. Depending on the fibre, it may represent contextuality, gauge mismatch, incompatible coarse-graining, insufficient boundary data, topology, or a defect in the chosen model. The first preprint does not identify all such failures with one metaphysical cause.

7. Reconstruction of Past and Future

> **Definition 7.1.** For present quasilocal data d_D at time t_0 and a model class M, let $\Gamma_D(d_D;M)$ be the set of histories compatible with the data, the domain structure, and the model.

$$\Gamma_D(d_D;M) = \{\gamma : \gamma \text{ is compatible with } d_D \text{ under } M \text{ in } D\}.$$

$$d_D \not\Rightarrow \gamma_\mathrm{unique}, \quad \text{but} \quad d_D \mapsto \Gamma_D(d_D;M).$$

A probabilistic reconstruction replaces the compatible set by a posterior measure:

$$P(\gamma_{<t_0} \mid d_D, M), \quad P(\gamma_{>t_0} \mid d_D, M).$$

> **Definition 7.2.** A quantity $Q(\gamma)$ is identifiable from d_D under M when Q is constant, or sufficiently concentrated within a declared tolerance, over $\Gamma_D(d_D;M)$.

> **Definition 7.3.** Structural uncertainty is dependence of the reconstruction on the model class itself, rather than only on parameters within one fixed model.

Past and future are thus reconstructed continuations, not directly observed objects. A singularity or Planck-scale boundary must be classified by the failure mode of the reconstruction or dynamical model: geodesic incompleteness, coordinate divergence, effective-theory breakdown, loss of identifiability, or genuine limiting behaviour.

8. The B^4 Filling as a Worked Example

Let B^4_A be the round four-ball with boundary S^3_A. Its use is auxiliary unless an independent physical postulate promotes it.

$$\partial B^4_A = S^3_A, \quad V_4(A) = \pi^2 A^4/2, \quad V_4/V_3 = A/4.$$

Given a homogeneous filling density $\rho_4(A)$, define

$$E_4(A) = \rho_4(A) V_4(A).$$

The induced boundary density and adiabatic pressure are

$$\varepsilon_D(A) = E_4(A)/V_3(A) = A\rho_4(A)/4,$$

$$p_D(A) = -dE_4/dV_3 = -\varepsilon_D(A) - (A/3)d\varepsilon_D/dA.$$

Within the homogeneous homothetic class,

$$\rho_4(A) = 4\varepsilon_D(A)/A.$$

This map is exact as a representation, but it is not yet predictive: every separately conserved homogeneous density can be represented in this form. Predictivity begins only when E_4 or ρ_4 is selected independently by an action, symmetry, spectral condition, compensation principle, or internal dynamics.

> **RTE reading 8.1.** B^4 may serve as a geometric filling, variational support, generator of boundary functionals, or carrier of reconstruction. None of these roles, by itself, licenses the claim that B^4 is an observed physical exterior.

8.1 Boundary functionals

$$\Pi^{ab} = \delta W/\delta h_ab, \quad J_A = \delta W/\delta \varphi^A.$$

Any dependence on the choice of filling must be classified as physical, gauge, topological, or evidence that the boundary description is incomplete.

9. Observer as Physical Subsystem

$$O = (H_O, \mathcal{A}_O, I_O, \tau_O).$$

> **Definition 9.1.** An observer structure is a stable physical subsystem capable of maintaining distinguishable internal states, coupling to signals, and ordering records sufficiently to support reconstruction.

This definition does not require consciousness. Detectors, memory-bearing systems, biological organisms, and engineered apparatuses may realize different observer structures. A biological observer is a special case.

$$E_\{D,O,M\} : S_phys \to S_obs(D,O,M).$$

Observer dependence is parameter dependence, not arbitrariness. Once D, O, M, admissible operations, and tolerance are fixed, the equivalence classes and reconstruction limits are determinate.

> **Proposition 9.2.** The disappearance of one observer structure changes the expression maps involving that observer; it does not imply disappearance of physical reality.

10. Distinguishability and Information

RTE treats information as downstream from physically realizable distinction. Before a Shannon measure can be assigned, a carrier must support distinguishable states under declared operations and resolution.

> **Definition 10.1.** The expression capacity of a domain-interface pair (D,O) is the physically admissible distinguishability profile that can be carried, causally integrated, and reconstructed by that pair at a declared resolution.

$$D\#_{q,\Lambda}(X) = (D\#_{top,\leq q}(X), C(X), Spec_{\leq\Lambda}(X)).$$

A mathematical mode count is not automatically a bit capacity. A finite alphabet, energy or occupation cutoff, code subspace, or gravitational selector is required before a bit interpretation is licensed.

Loss of distinction must be attributed to an explicit morphism: quotient, deletion, coefficient change, structure forgetting, spectral truncation, or coarse-graining.

$$\ker(E_D) \text{ records distinctions unavailable to the domain.}$$

In this sense, ignorance is not a vague absence. It appears as a kernel, equivalence relation, obstruction, or posterior spread.

11. Perceptual Example: Colour as Relational Manifestation

Let X be an external physical excitation, S a sensory system, N neural dynamics, C conscious access, K context and memory, and Q the qualified content “blue”.

$$Q = F(X, S, N, C, K).$$

The manifestation is not merely X and not merely C. It is a jointly determined structure

$$P_D = (X, S, N, C, Q)_D.$$

> **Interpretation 11.1.** Blue is neither a property of an isolated object nor a little object located inside consciousness. It is a qualified manifestation of the complete physical-perceptual configuration.

This example is illustrative, not a theory of consciousness. Its role is to show that instrument dependence does not imply unreality; it implies that the observable belongs to the full measurement configuration.

12. Invariants and Model Adequacy

> **Definition 12.1.** An RTE invariant is a quantity, relation, or structure preserved under a declared class of admissible changes of domain, observer interface, reconstruction procedure, or formal representation.

The admissible class must be specified; otherwise the word invariant has no content.

Criterion	Requirement
Empirical accuracy	Predicted manifestations agree with data within a stated tolerance.
Invariance	Core relations survive declared admissible changes.
Stability	Small perturbations do not destroy the conclusion.
Transferability	Descriptions agree across nested or overlapping domains.
Minimal excess structure	Auxiliary mathematics is not promoted to untestable ontology.
Identifiability	Important quantities depend on quotient classes, not arbitrary representatives.
Boundary discipline	Boundary and filling terms are classified explicitly.
Falsifiability	A possible observation or reconstruction failure can reject the postulate.

$$M_1 \simeq_{\{\mathcal{D},\mathcal{O},\varepsilon\}} M_2$$

means that, after quotienting gauge and representation redundancy, the two models yield the same accessible manifestations within tolerance ε for all admitted domains, observers, and candidate states.

Agreement at homogeneous background level does not imply full physical equivalence.

13. Minimal Theorem Package

Version 1.0 deliberately proves only what the present definitions support.

Result	Status	Purpose
Operational equivalence	Theorem 5.3	Defines domain objects as equivalence classes.
Quotient factorization	Proposition 5.5	Separates inaccessible representatives from accessible image.
Sheaf gluing condition	Proposition 6.4	Separates local compatibility from global realizability.
Typed B^4 discipline	Proposition 3.2	Prevents illicit promotion of an auxiliary filling.
Compatible histories	Definitions 7.1–7.3	Separates observation, retrodiction, prediction, and model dependence.

13.1 Claims excluded from Version 1.0

- RTE does not derive physical law from ontology.
- B^4 is not asserted to be a physical exterior.
- The expression relation is not assumed to be functorial before its categories are fixed.
- Consciousness is not required for physical expression.
- A reconstruction boundary is not automatically a physical singularity.
- Mode counts are not bit counts without an operational selector.
- Cohomological obstruction is not claimed until a concrete complex and coefficient system are specified.

14. Empirical Contact and Falsifiability

> **Falsifier 14.1.** If an RTE construction yields no restriction beyond ordinary operational equivalence, sheaf consistency, or Bayesian inverse modelling, then it is a vocabulary rather than a new physical theory.
>
> **Falsifier 14.2.** If a B^4 residual sector cannot supply tensorial completion, exchange law, perturbation theory, and an observational map beyond homogeneous reconstruction, it remains a background representation rather than a completed physical model.
>
> **Falsifier 14.3.** If an alleged invariant changes under a declared admissible transformation, then either it is not invariant or the admissible class was defined incorrectly.

For the cosmological branch, empirical contact requires the full chain

$$E_4(A,\xi) \to (\varepsilon_D, p_D, Q_D) \to T^D_{\mu\nu} \to \text{background} + \text{perturbations} \to \text{observables}.$$

At each arrow the status must be stated: proved, postulated, fitted, or open.

15. Research Programme

- Paper I: RTE as quasilocal operational equivalence and typed reconstruction — the present work.
- Paper II: Sheaf and stack formulations of quasilocal expression on compact S^3.
- Paper III: B^4 filling functionals, filling independence, tensorial completion, and perturbations.
- Paper IV: Observer structures and carrier-dependent distinguishability.
- Paper V: Empirical model comparison, robustness, and observational falsifiers.

The order matters. Without the present paper the ontology remains diffuse. Without the local-to-global paper, quasilocality remains geometric but not logically controlled. Without the variational paper, the B^4 sector remains a representation. Without the observer and information paper, the interface remains a slogan. Without empirical comparison, the programme has no physical discipline.

Conclusion

The first motif can now be stated in exact form. Expression is represented by domain-indexed relations or maps. These induce operational equivalence classes. Local expression is organized by restriction and gluing. Reconstruction yields compatible histories rather than direct possession of unobserved being. Auxiliary mathematical objects are kept within their proper types. Information enters only after physically realizable distinction has been defined.

$$E_D \text{ induces } \sim_D; \quad \sim_D \text{ induces } [x]_D;$$

$$\text{gluing tests local-to-global expression;}$$

$$\text{reconstruction yields } \Gamma_D(d_D;M), \text{ not direct observation of the past or future.}$$

This is enough for Version 1.0. It fixes the motif without pretending that the entire composition is already complete.

Appendix A. Axioms and Postulates

> **Axiom A1 — Local accessibility.** Every physically accessible content is indexed by a causal-geometric domain and an admissible expression structure.
> **Axiom A2 — Operational equivalence.** States indistinguishable by all admitted expressions in D define one domain object.
> **Axiom A3 — Restriction compatibility.** Domain restriction preserves declared expression equivalence whenever the restriction maps are defined.
> **Axiom A4 — Reconstruction limitation.** Present data determine a compatible class or posterior of histories, not necessarily one unique history.
> **Axiom A5 — Type discipline.** Mathematical existence does not entail physical existence without an additional physical postulate.
> **Postulate P1 — Physical interface.** Every operational expression map is instantiated by a physically admissible observer or measurement interface.
> **Postulate P2 — Empirical closure.** A physical branch of RTE must terminate in an observational map with stated tolerances and failure conditions.

Appendix B. Formula Register

Concept	Expression
Typed data	R, C_D, O, M, P_D
Expression relation	$\mathcal{E}_D \subset S_{phys} \times S_{obs}(D)$
Expression map	$E_D : S_{phys} \to S_{obs}(D)$
Operational equivalence	$x_1 \sim_D x_2 \Leftrightarrow E_D(x_1)=E_D(x_2)$
Domain object	$[x]_D = E_D^{-1}(E_D(x))$
History space	$\Gamma_D(d_D;M)$
Presheaf restriction	$\rho_{DU} : \mathcal{F}(D) \to \mathcal{F}(U)$
Gluing	$s_i\|_{D_i \cap D_j}=s_j\|_{D_i \cap D_j}$
Round S^3 volume	$V_3(A)=2\pi^2A^3$
Round B^4 volume	$V_4(A)=\pi^2A^4/2$
Boundary density	$\varepsilon_D=A\rho_4/4$
Pressure	$p_D=-\varepsilon_D-(A/3)d\varepsilon_D/dA$
Observer tuple	$O=(H_O,\mathcal{A}_O,I_O,\tau_O)$

Appendix C. Glossary

Term	Definition
Expression	Domain-conditioned availability of physical content through causal-geometric, observer, operational, and formal structures.
Expressibility	The primitive compatibility structure under which a manifestation can occur.
Manifestation	Accessible content P_D licensed in a quasilocal domain.
Ground candidate	A placeholder for compatible underlying realizations; not a hidden classical object by default.
Quasilocal domain	A physically distinguishable region with finite causal and observational access.
Operational equivalence	Equality of all admitted manifestations or outcome statistics.
Domain object	The quotient class [x]_D accessible through E_D.
Reconstruction	A constrained continuation from present data to compatible histories.
Observer structure	A physical subsystem carrying distinguishable states, interfaces, and internal ordering.
Auxiliary filling	A mathematical construction used to organize boundary quantities without automatic physical promotion.
Expression obstruction	Failure of local data to admit a global realization under declared gluing rules.
Expression invariant	A structure preserved under a declared class of admissible changes.

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